

Deepak R V

✉ deepakvetrivelan@gmail.com ☎ +91 78452 10074 in linkedin.com/in/deepakrv07 🐙 github.com/DEEPAKRV07

Professional Summary

AI/ML-focused Computer Science and Engineering student with hands-on experience in computer vision, mobile application development, embedded systems, and intelligent automation. Built real-world projects using Python, Flutter, OpenCV, TensorFlow Lite, and YOLOv8 with a focus on accessibility, sports analytics, and practical problem solving.

Education

S.A. Engineering College, Chennai

2022 – 2026

Bachelor of Engineering in Computer Science and Engineering (AI/ML Specialization)

CGPA: 8/10

- **Activities & Societies:** IEEE Student Member, coordinated a department-level hackathon, participated in technical events and project exhibitions.
- **Final-Year Project:** SightMate – AI Navigation Assistant for the Visually Impaired.
- **Coursework:** Data Structures, Algorithms, Database Management Systems, Machine Learning, Artificial Intelligence, Computer Vision.

Experience

Account Manager & Designer – The Quality Threads

Jan 2024 – Mar 2025

- Managed client communication, quotations, order tracking, and production coordination.
- Designed branding assets, promotional materials, and social media graphics using Canva and Adobe Photoshop.
- Maintained Excel-based tracking and documentation workflows for orders and deliveries.
- Coordinated with clients and production teams to support timely execution of work.

Intern – Embedded Systems, MSME, Guindy, Chennai

Aug 2024 – Sep 2024

- Developed automation projects using Arduino boards and IoT components.
- Programmed sensors and output modules for real-time embedded control systems.
- Worked on microcontroller integration, circuit debugging, and prototype testing.
- Gained practical exposure to embedded hardware workflows and system troubleshooting.

Projects

SightMate – AI Navigation Assistant for the Visually Impaired

github.com/DEEPAKRV07/SightMate-AI-Assistant

- Developed a Flutter-based mobile application that assists visually impaired users through real-time navigation and voice interaction.
- Integrated YOLOv8 TensorFlow Lite for real-time object detection and Fast-SCNN semantic segmentation to detect obstacles and walkable paths.
- Designed a navigation fusion pipeline that combines detection and segmentation outputs to generate movement guidance such as move left, move right, and path clear.
- Implemented OCR using Google ML Kit for reading printed text and Text-to-Speech for hands-free audio feedback.
- Added accessibility features including voice commands, Braille touch keyboard input, speech translation, and system information announcements.
- Built a fully offline AI inference pipeline with modular camera, detection, OCR, navigation, translation, and speech services.
- *Tech Stack:* Flutter (Dart), TensorFlow Lite, YOLOv8, Fast-SCNN, Google ML Kit, Speech-to-Text, Text-to-Speech, Android Studio, Git

Football Match Video Analysis using YOLOv8 & ByteTrack

github.com/DEEPAKRV07/Football-Analysis-System

- Developed an AI-based football analytics system to detect and track players, referees, and the ball from match videos using YOLOv8 and ByteTrack.
- Implemented camera motion estimation and view transformation to improve player

position tracking on the field.

- Calculated player speed, distance covered, and ball possession using tracking data and frame-by-frame analysis.
- Applied K-Means clustering for automatic team classification based on jersey color detection.
- Designed a full video processing pipeline including detection, tracking, team assignment, and statistical analysis.
- Generated annotated output videos with player bounding boxes, IDs, team labels, ball markers, and tracking visualization.
- *Tech Stack: Python, OpenCV, YOLOv8, ByteTrack, Scikit-learn, NumPy, Pandas*

Kaatchi Media – Frontend Web Development

kaatchimedia.com

- Collaborated in the development, optimization, testing, and production deployment of the official Kaatchi Media portfolio website using HTML5, CSS3, and modern JavaScript (ES6).
- Refactored the frontend into a modular component-based architecture with reusable navigation, gallery, animations, and UI components for improved maintainability and scalability.
- Developed responsive layouts, interactive photography galleries, testimonial carousel, portfolio filtering, smooth animations, and production-ready Web3Forms contact integration.
- Optimized frontend performance through lazy loading, responsive image handling, reusable JavaScript modules, cross-browser testing, and Git-based collaborative development workflows.
- *Tech Stack: HTML5, CSS3, JavaScript (ES6), CSS Grid, Flexbox, Web3Forms, Git, GitHub*

Virtual Mouse using Hand Gestures

github.com/DEEPAKR07/Virtual-Mouse-Controll

- Developed a webcam-based virtual mouse system using hand gesture recognition and real-time computer vision.
- Implemented hand landmark detection using MediaPipe to track finger positions and interpret gestures.
- Designed gesture logic for cursor movement, left click, right click, double click, and screenshot actions.
- Optimized real-time processing for smoother cursor control and low-latency user interaction.
- *Tech Stack: Python, OpenCV, MediaPipe, PyAutoGUI, pynput, NumPy*

replace content

Technical Skills

Languages: Python, SQL, HTML5, CSS3, JavaScript

Frameworks/Libraries: Django, Flutter, Google ML Kit, Scikit-learn, Pandas, NumPy

AI & Computer Vision: YOLOv8, Fast-SCNN, OpenCV, TensorFlow Lite, MediaPipe, OCR, Object Detection, Semantic Segmentation, Multi-Object Tracking, K-Means Clustering, Machine Learning, Deep Learning

Databases: SQLite3, SQL Server

Tools: Git, GitHub, Android Studio, Power BI, Canva,

Certifications

MSME Technology Development Centre (PPDC) – Embedded Systems Internship Certification

NPTEL – Privacy and Security in Online Social Media